Exercise – Understanding the theory of block bootstrap

1. Aim of the exercise

Under standard bootstrap theory, observations are assumed to be independent and identically distributed (i.i.d.). In time series data, however, observations that are close in time are typically correlated, violating this assumption. As a result, the standard bootstrap, which resamples individual observations, generally fails to reproduce the dependence structure of the data. The block bootstrap addresses this problem by resampling blocks of consecutive observations, thereby preserving local dependence. Here we show that the block bootstrap restores consistency and that this consistency requires suitable choices of the block length, a tuning parameter selected by the researcher.

2. Asymptotic behavior of the sample mean

Let $X_1, \ldots, X_n$ be a sequence of observations from a strictly stationary time series process with constant mean μ and finite variance γ_0. Strict stationarity dictates that the joint probability distribution of a block of k consecutive observations does not depend on its time origin. Formally, for any window length $k \in \mathbb{N}_+$ and time shift t, the baseline block and the shifted block share the identical joint distribution:

$$(X_1, X_2, \ldots, X_k) \stackrel{d}{=} (X_t, X_{t+1}, \ldots, X_{t+k-1})$$

Consequently, the temporal dependence structure depends entirely on the relative distance or lag h between observations, rather than the absolute time t. We measure this time-invariant relationship via the autocovariance function:

$$\gamma_h = \mathrm{Cov}(X_t, X_{t+h}), \qquad h = 0, \pm 1, \pm 2, \ldots$$

For example, under strict stationarity, a 3-day market snapshot ($k = 3$) starting on Monday (1) follows the exact same probability distribution as a 3-day snapshot shifted to start two weeks later ($t = 15$, spanning time steps $15, 16$, and 17).

Consider the sample mean as a statistic, given by

$$T_n = \sqrt{n}(\bar{X}_n - \mu) = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} (X_t - \mu).$$

The variance of T_n is

$$v_n = \mathrm{Var}(T_n) = \frac{1}{n} \mathrm{Var}\left(\sum_{t=1}^{n} (X_t - \mu) \right).$$

Assume time series data to study the behavior of the sample mean. For i.i.d. data, only the term $h = 0$ contributes. To see this, expand the variance of the sum as a double sum of

covariances:

$$\begin{aligned} v_n &= \frac{1}{n}\operatorname{Var}\left(\sum_{t=1}^{n}(X_t-\mu)\right) \\ &= \frac{1}{n}\sum_{t=1}^{n}\sum_{s=1}^{n}\operatorname{Cov}(X_t-\mu, X_s-\mu) \\ &= \frac{1}{n}\sum_{t=1}^{n}\sum_{s=1}^{n}\operatorname{Cov}(X_t, X_s). \end{aligned}$$

Since the observations are independent,

$$\operatorname{Cov}(X_t, X_s) = 0, \qquad t \neq s,$$

and

$$\operatorname{Cov}(X_t, X_t) = \gamma_0.$$

Therefore,

$$\begin{aligned} v_n &= \frac{1}{n}\sum_{t=1}^{n}\gamma_0 && (1) \\ &= \gamma_0 && (2) \\ &= \sigma^2. \end{aligned}$$

Under time series dependence, however, the variance of T_n depends on the entire autocovariance structure of the process. If the autocovariances are absolutely summable, then the variance of T_n converges to

$$v_n \to \sigma_\infty^2 = \sum_{h=-\infty}^{\infty}\gamma_h = \gamma_0 + 2\sum_{h=1}^{\infty}\gamma_h,$$

where σ_∞^2 is known as the long-run variance. The long-run variance therefore incorporates the effect of all autocovariances rather than only the contemporaneous variance γ_0.

Hence, even asymptotically, the sampling distribution depends on the long-run variance σ_∞^2. With positive autocorrelation,

$$\sigma_\infty^2 > \gamma_0,$$

because successive observations contain overlapping information. Consequently, the effective amount of information in the sample is smaller than under independence, and the sample mean exhibits greater variability.

Hence, under standard weak-dependence conditions,

$$T_n = \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma_\infty^2).$$

The limiting distribution function of T_n takes the form

$$G_\infty(\tau, P) = \Phi\left(\frac{\tau}{\sigma_\infty}\right),$$

where P denotes the data-generating process.

3. Serial dependence and the failure of the standard bootstrap

We now examine whether the bootstrap method remains valid when data is not independent and identically distributed. This requires constructing a bootstrap analogue, T_n^*, whose conditional distribution asymptotically reproduces the limiting distribution of T_n established in the preceding section.

Under the standard bootstrap approach, we treat the observed sample as the population and draw a bootstrap sample $X_1^*, \ldots, X_n^*$ by resampling observations individually and with replacement from the original sequence $\{X_t\}_{t=1}^n$. As a result, the bootstrap observations are conditionally i.i.d. given the data. To evaluate the sample mean, we define the bootstrap statistic as:

$$T_n^* = \sqrt{n}(\bar{X}_n^* - \bar{X}_n),$$

where

$$\bar{X}_n^* = \frac{1}{n}\sum_{t=1}^{n} X_t^*$$

is the mean of the bootstrap sample. Conditioning on the observed sample, $\bar{X}_n$ is the mean of the empirical distribution from which the bootstrap observations are drawn.

This individual resampling mechanism introduces a fundamental flaw when applied to time series data. Because the limiting distribution of the real-world statistic T_n depends on the entire joint distribution P, any valid bootstrap must capture this joint dependence. However, drawing individual observations completely at random breaks the chronological order of the data. This breaks the chain of autocovariances, forcing the bootstrap world to act as if $\gamma_h = 0$ for all $h \neq 0$.

To see this, consider the conditional mechanics of the standard bootstrap. Given the observed sample, the bootstrap observations X_t^* are conditionally i.i.d. draws from the empirical distribution, where $P^*(X_t^* = X_i) = \frac{1}{n}$ for $i = 1, \ldots, n$. Let $\mathbb{E}^*[\cdot]$ and $\mathrm{Var}^*(\cdot)$ denote the conditional expectation and variance operators given the original data. By definition, the conditional moments are:

$$\mathbb{E}^*[X_t^*] = \sum_{i=1}^{n} X_i \, P^*(X_t^* = X_i) = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}_n,$$

and

$$\mathrm{Var}^*(X_t^*) = \mathbb{E}^*[(X_t^* - \mathbb{E}^*[X_t^*])^2] = \frac{1}{n}\sum_{t=1}^{n}(X_t - \bar{X}_n)^2 = \hat{\gamma}_0.$$

Because the bootstrap observations are conditionally independent, the cross-covariance terms between different draws are zero. Therefore, the conditional variance of the bootstrap sample mean scales exactly as it would under standard independent sampling:

$$\mathrm{Var}^*(\bar{X}_n^*) = \mathrm{Var}^*\left(\frac{1}{n}\sum_{t=1}^{n} X_t^*\right) = \frac{1}{n^2}\sum_{t=1}^{n} \mathrm{Var}^*(X_t^*) = \frac{\hat{\gamma}_0}{n}.$$

Evaluating the conditional variance of the bootstrap statistic $T_n^* = \sqrt{n}(\bar{X}_n^* - \bar{X}_n)$ yields:

$$\mathrm{Var}^*(T_n^*) = \mathrm{Var}^*\left(\sqrt{n}\bar{X}_n^*\right) = n \; \mathrm{Var}^*(\bar{X}_n^*) = n\left(\frac{\hat{\gamma}_0}{n}\right) = \hat{\gamma}_0.$$

This algebraic simplification demonstrates that the standard bootstrap variance collapses entirely to the contemporaneous sample variance, completely omitting the serial covariances that characterize the underlying dependent process.

We now discuss the asymptotic inconsistency for dependent data. By the Law of Large Numbers, the sample variance consistently estimates the contemporaneous variance as the sample size grows:

$$\hat{\gamma}_0 \xrightarrow{p} \gamma_0.$$

Applying a conditional Central Limit Theorem, the conditional distribution of T_n^* converges to a normal distribution scaled entirely by this single parameter:

$$T_n^* \mid X_1, \ldots, X_n \xrightarrow{d} \mathcal{N}(0, \gamma_0).$$

Recall, however, that the true target limiting distribution of the original time series statistic T_n is:

$$T_n \xrightarrow{d} \mathcal{N}(0, \sigma_\infty^2),$$

where

$$\sigma_\infty^2 = \gamma_0 + 2\sum_{h=1}^{\infty} \gamma_h.$$

Except in the highly restrictive, knife-edge scenario where the lagged autocovariances sum exactly to zero ($\sum_{h=1}^{\infty} \gamma_h = 0$), these two limiting distributions will differ.

By treating data points as independent entities, the standard bootstrap completely drops the time-dependent covariance structure (γ_h for $h \neq 0$). It forces the bootstrap world to assume that the past holds no memory over the future. As first noted by Singh (1981), the standard bootstrap is generally inconsistent for serially dependent data because it converges to the wrong limiting distribution, underestimating or overestimating the true variability (σ_∞^2) depending on the sign of the serial correlation.

4. The non-overlapping block bootstrap

The block bootstrap resamples blocks of consecutive observations to preserve local dependence. Choose a block length

$$l \in \{1, \ldots, n\},$$

and divide the sample into

$$k = \left\lfloor \frac{n}{l} \right\rfloor$$

full, non-overlapping blocks

$$Q_j = (X_{(j-1)l+1}, \ldots, X_{jl}), \qquad j = 1, \ldots, k. \tag{3}$$

where $k = \left\lfloor \frac{n}{l} \right\rfloor$ represents the number of full, non-overlapping blocks.

If n is not a multiple of l, the remaining $n - kl$ observations are discarded, leaving an effective sample size of $m = kl$ with sample mean $\bar{X}_m$.

A bootstrap sample

$$X_1^*, \ldots, X_m^*$$

is obtained by drawing k blocks with replacement from $Q_1, \ldots, Q_k$ and concatenating them. Observations within the same bootstrap block retain their original dependence structure, whereas observations belonging to different bootstrap blocks are independent.

Each block is drawn with probability $1/k$, so the bootstrap sample mean satisfies

$$\mathbb{E}^*(\bar{X}_m^*) = \bar{X}_m,$$

where $\mathbb{E}^*(\cdot)$ denotes expectation with respect to the resampling distribution. To study the distribution of the bootstrap sample mean, we consider the centred and scaled statistic

$$T_n^* = \sqrt{m}\,(\bar{X}_m^* - \bar{X}_m).$$

Its CDF given the sample,

$$G_n(\tau, P_l),$$

where P_l denotes the distribution generated by resampling blocks of length l, replaces the bootstrap CDF in Section 1 and is approximated using B bootstrap samples as in (8).

When $l = 1$, every block consists of a single observation, and the block bootstrap reduces to the standard bootstrap. The moving block bootstrap of Künsch (1989) uses all $n - l + 1$ overlapping blocks instead and is typically more accurate, although the role of the block length remains the same Lahiri (2003). We focus on the non-overlapping block bootstrap of Carlstein (1986) because it makes the role of the block length most transparent.

5. The block length and the conditions for consistency

The role of the block length is most evident in the variance of T_n^*. Let $\bar{Q}_j$ denote the mean of block j. Because a bootstrap sample consists of k blocks drawn independently and with equal probability, $\bar{X}_m^*$ is the average of k independently drawn block means from $\bar{Q}_1, \ldots, \bar{Q}_k$. The bootstrap variance of T_n^* is

$$\sigma_l^2 = \mathrm{Var}^*(T_n^*) = \frac{l}{k}\sum_{j=1}^{k}(\bar{Q}_j - \bar{X}_m)^2.$$

Each block is a stationary stretch of l observations, so

$$l\ \mathrm{Var}(\bar{Q}_j) = v_l,$$

where v_l denotes the variance of the centred and scaled sample mean based on a sample of size l. Because $\bar{X}_m$ is the average of the k block means, it follows that

$$\mathbb{E}(\sigma_l^2) = v_l - \frac{l}{m}v_m.$$

This expression reveals the two conditions that the block length must satisfy for the block bootstrap to be consistent.

First, the block length must increase with the sample size:

$$l \to \infty.$$

The quantity v_l contains only the autocovariances up to lag $l-1$, with weights $1 - h/l$, because the dependence between observations that end up in different blocks is broken when blocks are resampled independently. For a fixed block length, v_l therefore does not converge to σ_∞^2, and the bootstrap remains inconsistent no matter how large the sample becomes. When $l = 1$, the

expected bootstrap variance reduces to γ_0. As l increases, v_l approaches σ^2_∞, but only gradually. If

$$\sum_{h=1}^{\infty} h\gamma_h < \infty,$$

then the bias satisfies

$$v_l - \sigma^2_\infty \approx -\frac{2}{l}\sum_{h=1}^{\infty} h\gamma_h,$$

so that the bias is of order $1/l$.

Second, the number of blocks must also increase:

$$k \to \infty,$$

which is equivalent to

$$\frac{l}{n} \to 0.$$

Because the bootstrap variance is computed from the variation across the k block means, its variance is of order

$$\frac{1}{k} = \frac{l}{n}.$$

If l grows proportionally to n, then k remains fixed and $\frac{l}{m}v_m$ does not converge to zero. Consequently, σ^2_l does not converge to a constant, no matter how large the sample becomes. When the blocks are long, their means are approximately independent and normally distributed, and σ^2_l behaves approximately as

$$\sigma^2_\infty \frac{\chi^2_{k-1}}{k}.$$

The choice of l therefore involves a bias–variance trade-off. Short blocks fail to capture enough dependence, producing a bias of order $1/l$. Long blocks leave too few blocks available for resampling, producing a variance of order l/n. The first condition, $l \to \infty$, reduces the bias, whereas the second condition, $l/n \to 0$, reduces the variance. Balancing the squared bias against the variance implies that the mean squared error of the bootstrap variance is minimized when

$$l \propto n^{1/3}.$$

Hall et al. (1995) show that the optimal rate depends on the purpose of the bootstrap: $l \propto n^{1/3}$ for estimating a bias or variance, $l \propto n^{1/4}$ for a one-sided distribution function, and $l \propto n^{1/5}$ for a two-sided distribution function. The constant of proportionality depends on the unknown autocovariances, so in practice the block length is typically chosen using a rule of thumb or a data-driven procedure (e.g., Politis and White, 2004).

Formally, the block bootstrap for the sample mean is consistent if the process is strictly stationary, the dependence between observations decays sufficiently quickly as the lag increases (formalized through strong mixing conditions), a moment slightly higher than the second is finite, and the block length satisfies

$$l \to \infty \qquad \text{and} \qquad \frac{l}{n} \to 0.$$

Under these conditions (Kunsch, 1989; Lahiri, 2003),

$$\sup_{\tau\in\mathbb{R}} |G_n(\tau, P_l) - G_n(\tau, P)| \xrightarrow{p} 0, \qquad n \to \infty.$$

The consistency result therefore differs from that of the standard bootstrap in one important respect. The statistical error now depends not only on the sample size but also on the block length chosen by the researcher. It vanishes only if the block length grows with the sample size at a suitable rate. In contrast, the simulation error is unaffected and continues to be controlled by the number of bootstrap replications B.

6. Visualizing the convergence results

To illustrate how the statistical error of the block bootstrap depends on both the sample size and the block length, we conduct simulations using a stationary Gaussian AR(1) process,

$$X_t = \phi X_{t-1} + \varepsilon_t,$$

with $\phi = 0.7$ and i.i.d. innovations

$$\varepsilon_t \sim N(0, 1).$$

The process is initialized from its stationary distribution, so that $\mu = 0$. Its autocovariance function is

$$\gamma_h = \frac{\phi^h}{1 - \phi^2},$$

implying

$$\gamma_0 = 1.96.$$

The corresponding long-run variance is

$$\sigma_\infty^2 = \gamma_0 + 2\sum_{h=1}^{\infty} \gamma_h = \frac{1}{1-\phi^2} + \frac{2}{1-\phi^2}\sum_{h=1}^{\infty} \phi^h = \frac{1}{(1-\phi)^2} = 11.1.$$

The standard bootstrap therefore replaces the correct long-run variance,

$$\sigma_\infty^2 = 11.1,$$

with the much smaller variance

$$\gamma_0 = 1.96.$$

The dependence in this process decays geometrically, so the process satisfies the weak-dependence conditions required for block-bootstrap consistency. Because the process is Gaussian, $\bar{X}_n$ is exactly normally distributed and the sampling CDF is

$$G_n(\tau, P) = \Phi\left(\frac{\tau}{\sqrt{v_n}}\right),$$

where v_n denotes the variance of the centred and scaled sample mean under serial dependence. The bootstrap can therefore be compared directly with the exact sampling distribution.

We compare three block-length rules:

$$l = 1,$$

the standard bootstrap, which violates the condition

$$l \to \infty;$$

$$l = n^{1/3},$$

which satisfies both consistency conditions and attains the optimal rate for variance estimation; and

$$l = \frac{n}{5},$$

which violates the condition

$$\frac{l}{n} \to 0,$$

because it always uses

$$k = 5$$

blocks, regardless of the sample size.

The sample sizes are

$$n = 125,\ 1000,\ 8000,$$

so that both

$$n^{1/3}$$

and

$$\frac{n}{5}$$

are integers and no observations are discarded. For each sample size, we use the first n observations from a single simulated realization of the process. The number of bootstrap replications is fixed at

$$B = 10,000.$$

Figure 1 shows the bootstrap PDFs, with one row for each block-length rule and one column for each sample size. Figure 2 shows the corresponding CDFs. Because convergence in probability is a statement about behavior across repeated samples, a single realization may provide a misleading impression. Figure 3 therefore shows the ratio of the bootstrap variance to the true variance of the centred and scaled sample mean across 1,000 independent samples. The MATLAB code is provided at the end of this document.

With $l = 1$ (top row), the bootstrap distribution is much too narrow. Its variance is approximately 2.1 in every sample, close to $\gamma_0 = 1.96$, whereas the true variance is approximately 11. The largest distance between the bootstrap and sampling CDFs remains roughly constant at 0.19 for all three sample sizes (0.193, 0.190, and 0.197). Increasing the sample size therefore does not improve the approximation. Although the bootstrap distribution converges as n increases, it converges to a distribution with variance γ_0 rather than the correct long-run variance.

With $l = n^{1/3}$ (middle row), the bootstrap distribution approaches the sampling distribution as the sample size increases. The largest distance between the bootstrap and sampling CDFs falls from 0.056 for $n = 125$ to 0.029 for $n = 1000$ and to 0.015 for $n = 8000$. This improvement is exactly what the consistency result predicts when the block length increases with the sample size while remaining small relative to it.

With $l = n/5$ (bottom row), the bootstrap resamples only five blocks, so the bootstrap sample mean can take at most 126 different values, one for each possible draw of five blocks with replacement. Consequently, the estimated PDF is spiky and the estimated CDF is a step function. The largest distances (0.105, 0.261, and 0.082) do not decrease systematically with the sample size. In this case, the apparent accuracy of the bootstrap depends more on which five blocks happen to be observed than on the sample size itself.

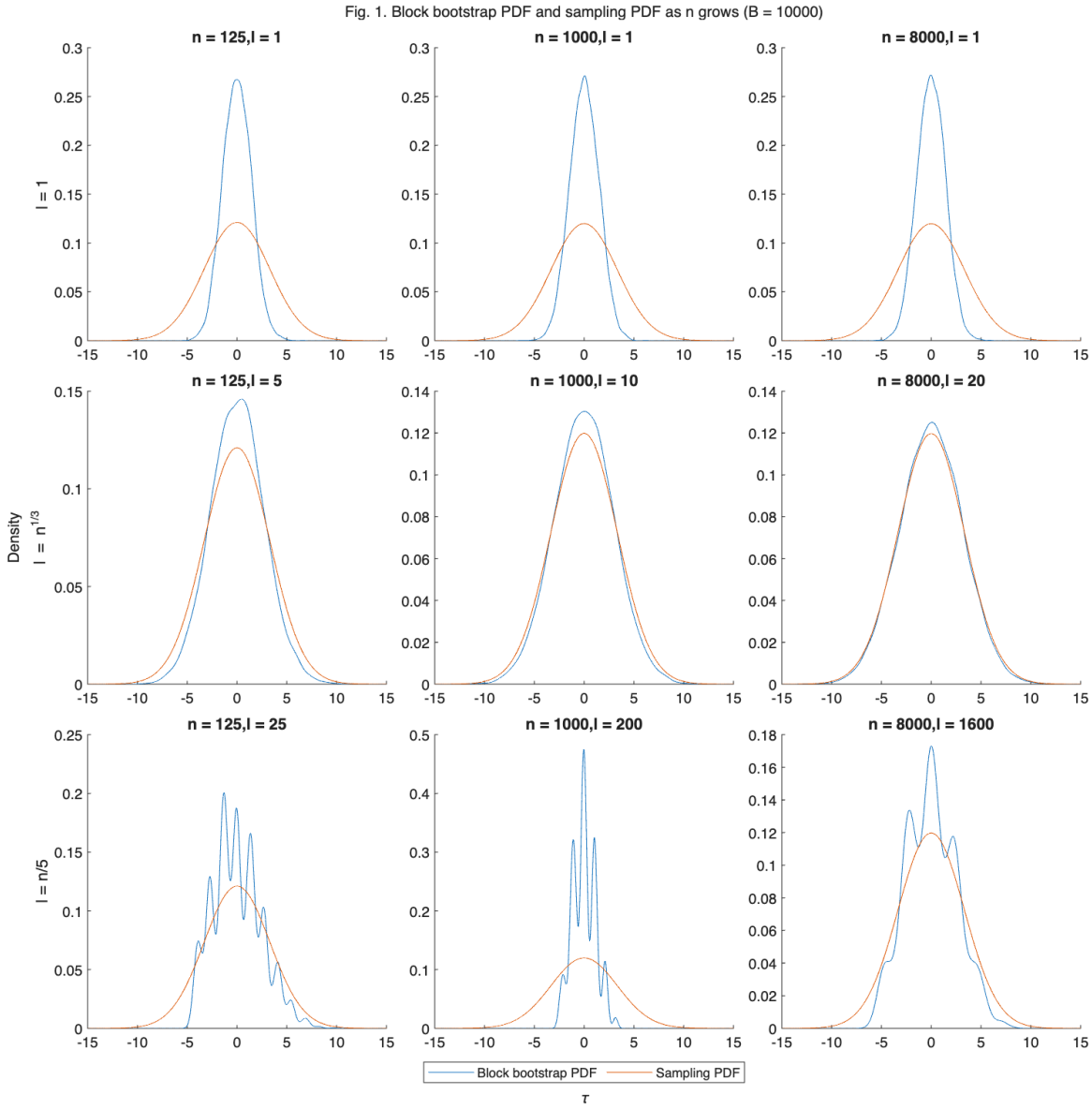


Figure 1: Block bootstrap PDFs and sampling PDFs for different sample sizes and block-length rules.

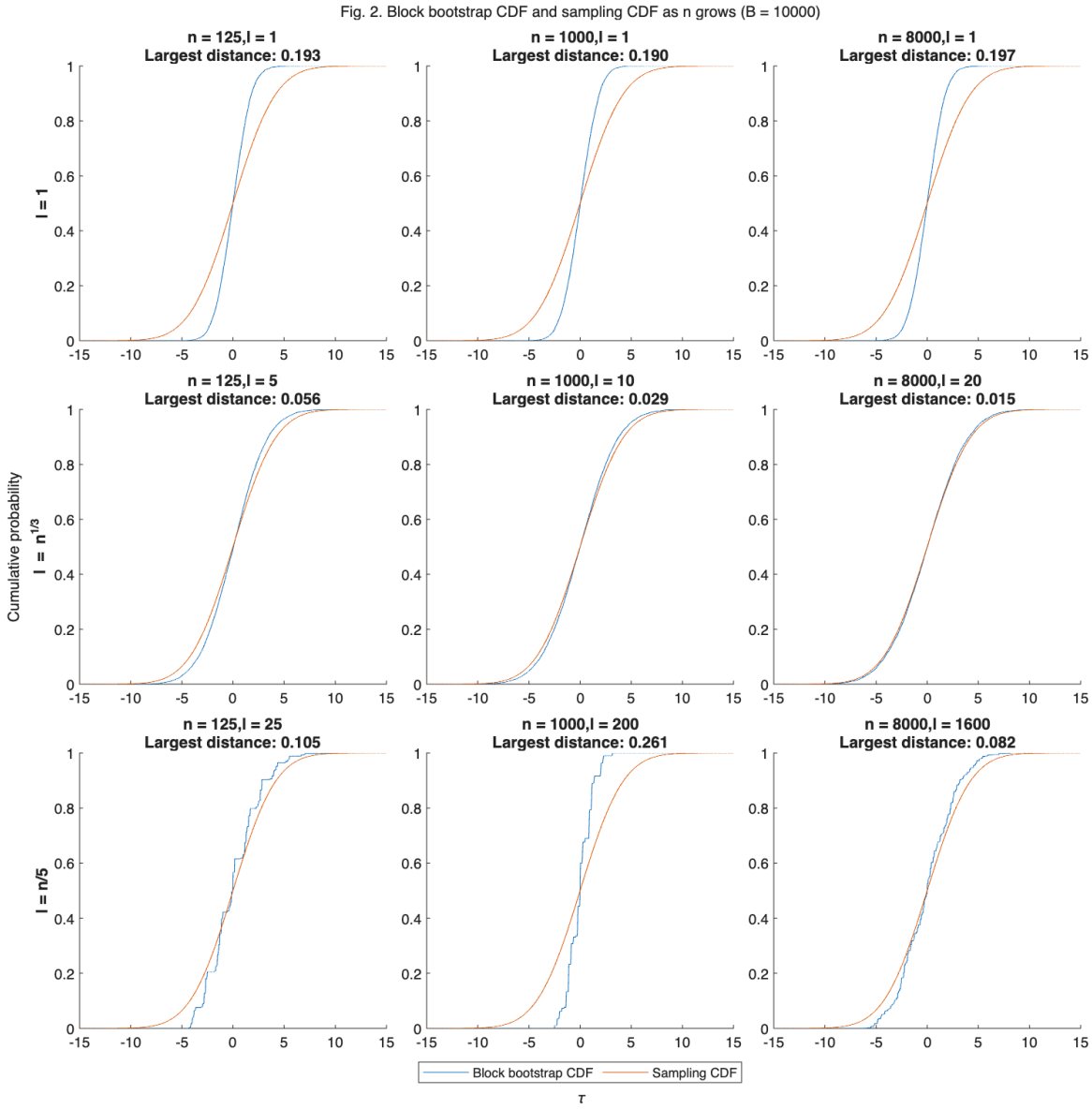


Figure 2: Block bootstrap CDFs and sampling CDFs for different sample sizes and block-length rules.

Figure 3 shows that these patterns are systematic. For $l = 1$, the ratio of the bootstrap variance to the true variance concentrates more and more tightly around

$$\frac{\gamma_0}{\sigma_\infty^2} = 0.18.$$

Its mean is 0.176 when $n = 8000$, and its standard deviation decreases from 0.038 to 0.005. The bootstrap therefore becomes increasingly precise about the wrong value.

For $l = n^{1/3}$, the distribution shifts toward one and becomes more concentrated. The means are 0.51, 0.73, and 0.86, while the standard deviations are 0.17, 0.11, and 0.06. These means agree closely with the theoretical values implied by the leading bias term of order $1/l$. Convergence is nevertheless slow. Even when $n = 8000$, the bootstrap still underestimates the variance by approximately 14% on average because the bias remains substantial when dependence is as persistent as in this process.

For $l = n/5$, the distribution changes little as the sample size increases. The mean remains between 0.69 and 0.80 and the standard deviation between 0.46 and 0.59, close to the values implied by the large-block approximation

$$\sigma_\infty^2 \frac{\chi_4^2}{5}.$$

With only a fixed number of blocks, increasing the sample size does not make the bootstrap more accurate.

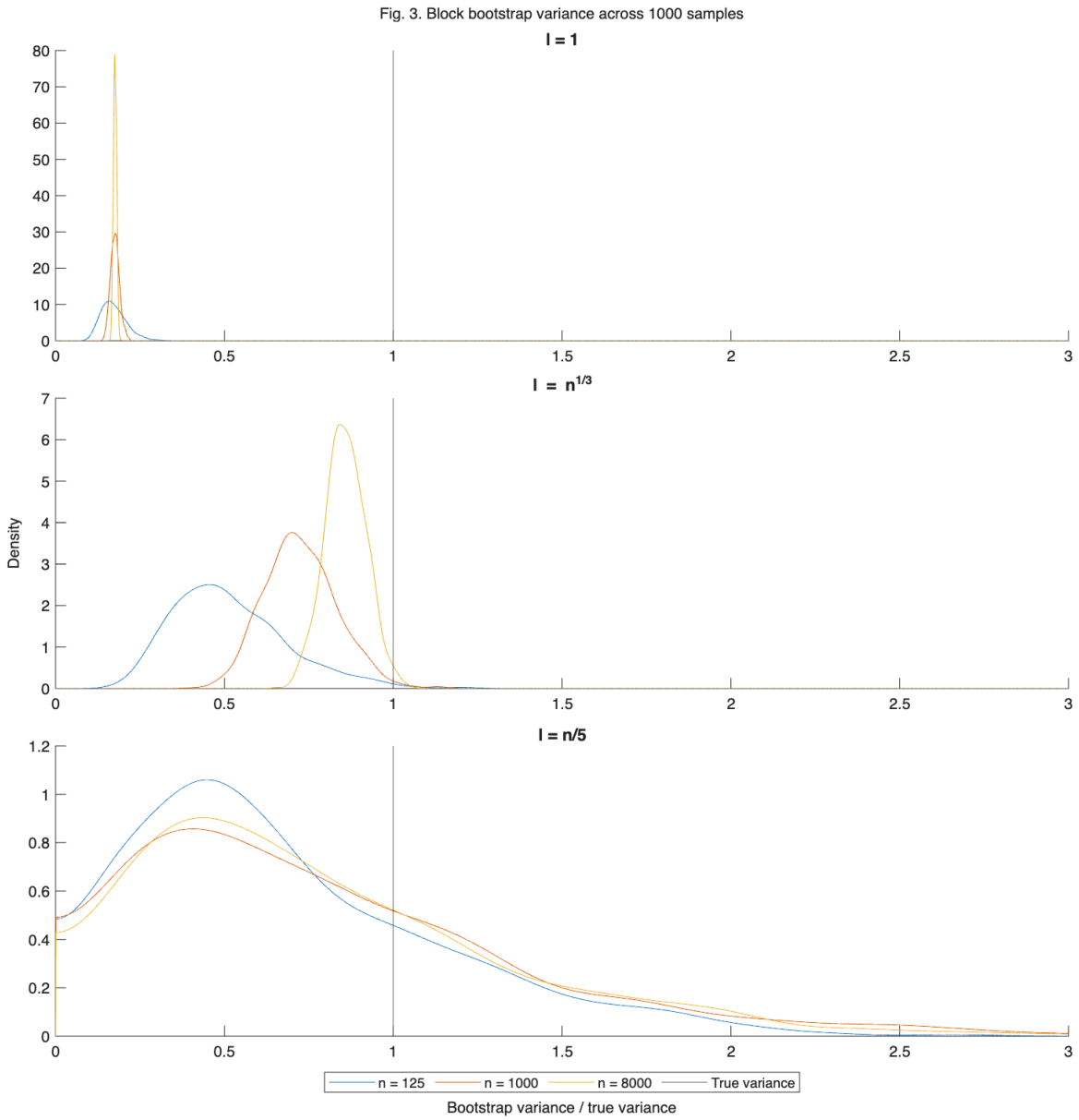


Figure 3: Distribution of the ratio of the bootstrap variance to the true variance across repeated samples.

The simulation therefore illustrates the trade-off faced by the researcher when choosing the block length. Blocks that are too short capture too little dependence and yield a bootstrap distribution that is precise but systematically incorrect. Blocks that are too long leave too few blocks for resampling and yield a bootstrap distribution that remains noisy regardless of the sample size. Even when the block length grows at the theoretically optimal rate, a large sample may still be required before the bootstrap approximation becomes accurate.

```
%% 3. Construct the data-generating process and analytical distributions

% 3.1. Set a seed for the random number generator for reproducibility
rng(26)

% 3.2. Set the autoregressive parameter of the AR(1) process
phi = 0.7;

% 3.3. Set the standard deviation of the innovations
sigma_eps = 1;

% 3.4. Compute the variance of the AR(1) process
gamma_0 = sigma_eps^2 / (1 - phi^2);

% 3.5. Define a function that simulates a stationary AR(1) series from a
% vector of innovations. The first observation is drawn from the
% stationary distribution, so that the series is stationary from the
% start.
ar1 = @(epsilon) filter(1,[1 -phi], ...
    [epsilon(1) / sqrt(1 - phi^2); epsilon(2:end)]);

% 3.6. Define the variance of the statistic for sample size n
v_pop = @(n) gamma_0 * (1 + 2 * sum((1 - (1:n-1) / n) .* phi.^(1:n-1)));

% 3.7. Define the sampling PDF of the statistic
g_pop = @(tau,n) pdf('Normal',tau,0,sqrt(v_pop(n)));

% 3.8. Define the sampling CDF of the statistic
G_pop = @(tau,n) cdf('Normal',tau,0,sqrt(v_pop(n)));

% 3.9. Set the grid of tau values at which PDFs and CDFs are evaluated
tau_grid = linspace(-15,15,1000)';

% 3.10. Set the sample sizes
N_obs_vec = [125 1000 8000];

% 3.11. Draw a sequence of observations from the AR(1) process
data_all = ar1(random('Normal',0,sigma_eps,[max(N_obs_vec),1]));

%% 4. Set the block-length rules

% 4.1. Define the block-length rules as functions of the sample size
block_rule = {@(n) 1, @(n) round(n^(1/3)), @(n) n / 5};
```

```
44
45 % 4.2. Set the labels of the block-length rules
46 block_label = {'l = 1', 'l = n^{1/3}', 'l = n/5'};
47
48 % 4.3. Set the number of block-length rules
49 N_rules = length(block_rule);
50
51 % 4.4. Set the sample sizes
52 N_n = length(N_obs_vec);
53
54 %% 5. Construct block bootstrap distributions for different n and l
55
56 % 5.1. Set the number of bootstrap replications
57 N_boot = 10^4;
58
59 % 5.2. Preallocate storage for the bootstrap statistic
60 T_boot = NaN(N_boot,N_rules,N_n);
61
62 % 5.3. For each block-length rule and sample size, select the original
63 % sample, divide it into non-overlapping blocks, draw bootstrap samples
64 % of blocks, and compute the bootstrap statistic
65 for j = 1:N_rules
66     for i = 1:N_n
67         n = N_obs_vec(i);
68         l = block_rule{j}(n);
69         k = n / l;
70         data_sample = data_all(1:n);
71         % Each row of blocks is one block of l consecutive observations
72         blocks = reshape(data_sample,l,k)';
73         for b = 1:N_boot
74             data_boot = datasample(blocks,k,1,'Replace',true);
75             % Tn* = sqrt(n) * (bootstrap sample mean -
76             % original sample mean)
77             T_boot(b,j,i) = sqrt(n) * ...
78                 (mean(data_boot(:)) - mean(data_sample));
79         end
80     end
81 end
82
83 %% 6. Convergence of the block bootstrap PDF to the sampling PDF
84
85 % Plot the block bootstrap PDF and the true sampling PDF for each
86 % block-length rule (rows) and sample size (columns).
87 figure('Position',[100 100 1000 1000]);
88 layout = tiledlayout(N_rules,N_n, ...
89     'TileSpacing','compact','Padding','compact');
90 for j = 1:N_rules
91     for i = 1:N_n
92         n = N_obs_vec(i);
93         l = block_rule{j}(n);
```

```
        nexttile
        hold on
        plot(tau_grid,ksdensity(T_boot(:,j,i),tau_grid, ...
            'function','pdf'))
        plot(tau_grid,g_pop(tau_grid,n))
        title(['n = ' num2str(n) ',l = ' num2str(l)])
        xlim([tau_grid(1) tau_grid(end)])
        if i == 1
            ylabel(block_label{j})
        end
        hold off
    end
end
x = xlabel(layout,'\tau');
y = ylabel(layout,'Density');
x.FontSize = get(groot,'defaultAxesFontSize');
y.FontSize = get(groot,'defaultAxesFontSize');
lgd = legend( ...
    'Block bootstrap PDF', ...
    'Sampling PDF', ...
    'Orientation','horizontal');
lgd.Layout.Tile = 'south';
t = title(layout,['Fig. 1. Block bootstrap PDF and sampling PDF ' ...
    'as n grows (B = ' num2str(N_boot) ')']);
t.FontSize = get(groot,'defaultAxesFontSize');
t.FontWeight = 'normal';

%% 7. Convergence of the block bootstrap CDF to the sampling CDF

% Plot the block bootstrap CDF and the true sampling CDF for each
% block-length rule (rows) and sample size (columns). The title of each
% panel reports the largest distance between the two CDFs over the grid
% of tau values.
dist_max = NaN(N_rules,N_n);
figure('Position',[100 100 1000 1000]);
layout = tiledlayout(N_rules,N_n, ...
    'TileSpacing','compact','Padding','compact');
for j = 1:N_rules
    for i = 1:N_n
        n = N_obs_vec(i);
        l = block_rule{j}(n);
        G_boot = arrayfun(@(t) mean(T_boot(:,j,i) <= t),tau_grid);
        dist_max(j,i) = max(abs(G_boot - G_pop(tau_grid,n)));
        nexttile
        hold on
        plot(tau_grid,G_boot)
        plot(tau_grid,G_pop(tau_grid,n))
        title({['n = ' num2str(n) ',l = ' num2str(l)], ...
            ['Largest distance: ' num2str(dist_max(j,i),'%.3f')]})
        xlim([tau_grid(1) tau_grid(end)])
```

```
        if i == 1
            ylabel(block_label{j},'FontWeight','bold')
        end
        hold off
    end
end
x = xlabel(layout,'\tau');
y = ylabel(layout,'Cumulative probability');
x.FontSize = get(groot,'defaultAxesFontSize');
y.FontSize = get(groot,'defaultAxesFontSize');
lgd = legend( ...
    'Block bootstrap CDF', ...
    'Sampling CDF', ...
    'Orientation','horizontal');
lgd.Layout.Tile = 'south';
t = title(layout,['Fig. 2. Block bootstrap CDF and sampling CDF ' ...
    'as n grows (B = ' num2str(N_boot) ')']);
t.FontSize = get(groot,'defaultAxesFontSize');
t.FontWeight = 'normal';

%% 8. Block bootstrap variance across repeated samples

% 8.1. Set the number of repeated samples
N_rep = 1000;

% 8.2. Preallocate storage for the ratio of the bootstrap variance to
% the true variance of Tn
var_ratio = NaN(N_rep,N_rules,N_n);

% 8.3. For each repeated sample and sample size, draw a new AR(1) series
% and compute the bootstrap variance for each block-length rule
for r = 1:N_rep
    for i = 1:N_n
        n = N_obs_vec(i);
        data_rep = ar1(random('Normal',0,sigma_eps,[n,1]));
        for j = 1:N_rules
            l = block_rule{j}(n);
            k = n / l;
            block_means = mean(reshape(data_rep,l,k),1)';
            var_boot = l * mean((block_means - mean(data_rep)).^2);
            var_ratio(r,j,i) = var_boot / v_pop(n);
        end
    end
end

% 8.4. Set the grid of ratio values at which the PDFs are evaluated
ratio_grid = linspace(0,3,3001)';

% 8.5. Plot the PDF of the variance ratio across repeated samples for
% each block-length rule and sample size. A ratio of 1 means that the
```

```
% bootstrap variance equals the true variance.
figure('Position',[100 100 1200 450]);
layout = tiledlayout(1,N_rules, ...
    'TileSpacing','compact','Padding','compact');
for j = 1:N_rules
    nexttile
    hold on
    for i = 1:N_n
        plot(ratio_grid,ksdensity(var_ratio(:,j,i),ratio_grid, ...
            'function','pdf','Support','positive', ...
            'BoundaryCorrection','reflection'))
    end
    xline(1);
    title(block_label{j})
    xlim([ratio_grid(1) ratio_grid(end)])
    hold off
end
x = xlabel(layout,'Bootstrap variance / true variance');
y = ylabel(layout,'Density');
x.FontSize = get(groot,'defaultAxesFontSize');
y.FontSize = get(groot,'defaultAxesFontSize');
lgd = legend( ...
    ['n = ' num2str(N_obs_vec(1))], ...
    ['n = ' num2str(N_obs_vec(2))], ...
    ['n = ' num2str(N_obs_vec(3))], ...
    'True variance','Orientation','horizontal');
lgd.Layout.Tile = 'south';
t = title(layout,['Fig. 3. Block bootstrap variance across ' ...
    num2str(N_rep) ' samples']);
t.FontSize = get(groot,'defaultAxesFontSize');
t.FontWeight = 'normal';

%% 9. Summarize the results

% Print, for each block-length rule and sample size, the block length,
% the number of blocks, the true variance of Tn, the bootstrap variance
% in the single sample of Figures 5 and 6, the largest distance between
% the CDFs, and the mean and standard deviation of the variance ratio
% across the repeated samples.
fprintf('\n%-12s %6s %6s %6s %8s %8s %8s %8s %8s\n','Rule', ...
    'n','l','k','v_n','Var*','Dist','MeanRat','SDRat')
for j = 1:N_rules
    for i = 1:N_n
        n = N_obs_vec(i);
        l = block_rule{j}(n);
        fprintf('%-12s %6d %6d %6d %8.3f %8.3f %8.3f %8.3f %8.3f\n', ...
            block_label{j},n,l,n/l,v_pop(n),var(T_boot(:,j,i)), ...
            dist_max(j,i),mean(var_ratio(:,j,i)),std(var_ratio(:,j,i)))
    end
end
```

```
fprintf('Variance of the AR(1) process (gamma_0): %.3f\n',gamma_0)
fprintf('Long-run variance sigma_eps^2/(1-phi)^2: %.3f\n', ...
    sigma_eps^2 / (1 - phi)^2)
```

7. Final notes

This file is prepared and copyrighted by Noortje Kamphuis, Axel Zoons and Tunga Kantarcı. This file and the accompanying MATLAB file are available on GitHub and can be accessed using this link. This file has made use of several references. First, Carlstein, E., 1986. The use of subseries values for estimating the variance of a general statistic from a stationary sequence. The Annals of Statistics, 14(3), 1171-1179. Second, Hall, P., Horowitz, J. L., Jing, B. Y., 1995. On blocking rules for the bootstrap with dependent data. Biometrika, 82(3), 561-574. Third, Künsch, H. R., 1989. The jackknife and the bootstrap for general stationary observations. The Annals of Statistics, 17(3), 1217-1241. Fourth, Lahiri, S. N., 2003. Resampling methods for dependent data. Springer. Fifth, Mignani and Rosa, 1995. The moving block bootstrap to assess the accuracy of statistical estimates in Ising model simulations. Computer physics communications. Sixth, Nordman and Lahiri, 2014. Convergence rates of empirical block length selectors for block bootstrap. Seventh, Radovanov and Marcikić, 2014. A comparison of four different block bootstrap methods. Croatian Operational Research Review. Eight, Singh, K., 1981. On the asymptotic accuracy of Efron's bootstrap. The Annals of Statistics, 9(6), 1187-1195. Ninth, Politis, D. N., White, H., 2004. Automatic block-length selection for the dependent bootstrap. Econometric Reviews, 23(1), 53-70.